MS – MATHEMATICS FORM II – 2026

1. (a) Importance of Mathematics:

i) In business and trade is used to calculate profit, loss, prices (01)

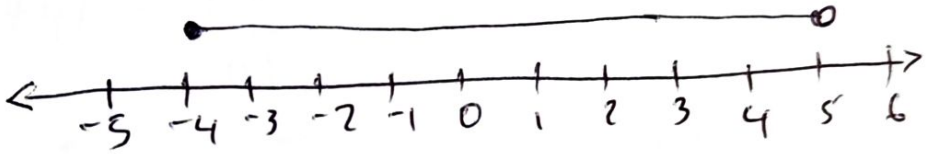
ii) In construction is used for designing and measurement (01)

iii) In transportation is used to calculate, distance, speed and fuel used (01)

Example: Shopkeeper calculating the total price of goods (01)

(*) Any relevant importance should be considered.

(b) (i) $-4 \leq x < 5$



(03)

(ii) submarine distance $|-200|$

aeroplane distance 200

total $|-200| + 200$

$= 200 + 200$

$= 400\text{m}$.

(03)

2 (a) Let $x = 0.\dot{4}4\dot{1}$

$1000x = 441.\dot{4}4\dot{1}$ ... (01)

$1000x = 441.\dot{4}4\dot{1}$

$-\ x \quad = \quad 0.\dot{4}4\dot{1}$

$999x = 441.$

$x = \frac{441}{999} \quad \frac{\cancel{147}}{\cancel{333}} \quad \frac{49}{111}$ ... (01)

$\therefore 0.\overline{441} = \frac{49}{111}$ ... (01)

In ratio for $0.\overline{441} = 49:111.$

(b) Pipe A fills the tank in 6 hrs; It fills $\frac{1}{6}$ of tank in 1 hr. (01)

Pipe B fills the tank in 4 hrs; It fills $\frac{1}{4}$ of tank in 1 hr.

Combined rate = Rate of Pipe A + Rate of Pipe B

$= \frac{1}{6} + \frac{1}{4}$

$= \frac{5}{12}$ ... (01)

If $\frac{5}{12}$ of tank is filled in 1 hr

$\frac{12}{12}$ will be filled in ? ... (01)

$\frac{5}{12}x = \frac{12}{12}$

$x = \frac{12}{5}$ (01)

$\therefore$ It will take 2 hrs and 24 minutes to fill tank.

4 (a) (i) Consider $\triangle BCF$ and $\triangle BCE$

$\overline{BF} = \overline{CE}$ - given ..... (0½)

$\overline{BC} = \overline{BC}$ - Common

$B\hat{F}C = C\hat{E}B$ - given ..... (0½)

$\therefore \triangle BCF \cong \triangle BCE$ by RHS.

$\therefore \overline{CF} = \overline{BE}$ - proved. ..... (01)

(ii) Since $\overline{ZY} // \overline{MN}$ - (they are both ⊥ar to $\overline{XN}$,).

it follows that $X\hat{Y}Z = X\hat{M}N$ - Corr. angles

$X\hat{Z}Y = X\hat{N}M$ - -"- } ..... (01)

Hence $\triangle XYZ \sim \triangle XMN$ (by A A Similarity Theorem)

Thus $\frac{\overline{XY}}{\overline{XM}} = \frac{\overline{XZ}}{\overline{XN}} = \frac{\overline{YZ}}{\overline{MN}}$

$\frac{10}{\overline{XM}} = \frac{9}{12} = \frac{6}{\overline{MN}}$ ..... (01)

$\overline{XM} = \frac{120}{9} = \frac{40}{3}$

$\overline{MY} = \overline{XM} - \overline{XY}$

$= \frac{40}{3} - 10$

$= \frac{10}{3}$ cm. ..... (01)

$\overline{MN} = \frac{12 \times 6}{9}$

$= 8$ cm ..... (01)

$\overline{MY} : \overline{MN} = \frac{10}{3} : 8 = 10 : 24$

4. (b)(i) $4 * (2 * y) = 4 \rightarrow$ Rule $a^2 - b$

$2 * y = 4 - y$

$4 * (4 - y) = 4$

$16 - (4 - y) = 4$ } ---- (01)

$16 - 4 + y = 4$

$y = 4 - 12$ ---- (01)

$y = -8.$

(ii) $T = 2\pi\sqrt{\frac{l}{g}}$

$\frac{T}{2\pi} = \sqrt{\frac{l}{g}}$ } ---- (01)

$\frac{T^2}{4\pi^2} = \frac{l}{g}$

$g = \frac{4\pi^2 l}{T^2}$ ---- (01)

5. (a) (i) $|6 - x| = 1$

$6 - x = 1$ or $-6 + x = 1$ ... (01)

$x = 5$ or $x = 7$. ... (01)

(ii) Let chickens be x and goats be y. 00½

$$\begin{cases} x + y = 10 & \text{.... (i)} \\ 4000x + 8000y = 60000 & \text{... (ii)} \end{cases}$$ ... 01

From (i) $x = 10 - y$ ... (iii) 00½

∴ Inserting (iii) into simplified (ii) we have

$4(10 - y) + 8y = 60$

$40 - 4y + 8y = 60$

$40 + 4y = 60$

$4y = 20$ 01

$y = 5$

$x = 10 - y$

$= 10 - 5$

$= 5$ 01

∴ The farmer bought 5 chickens and 5 goats.

5 (b) (i) $a^2 - b^2 = (a+b)(a-b)$ 00½

$(85.18)^2 - (14.82)^2 = (85.18 + 14.82)(85.18 - 14.82)$ 00½

$(100)\ (7036)$ 01

$7036.$

(ii) $5x^2 - 3x = 5\left(x^2 - \frac{3}{5}x + \left(\frac{3}{10}\right)^2\right)$ 00½

$= 5\left(x^2 - \frac{3}{5}x + \frac{9}{100}\right)$ 00½

$= 5x^2 - 3x + \frac{9}{20}$ 01

Alternatively:

$b^2 = 4ac \rightarrow c = \frac{b^2}{4a}$

$a = 5 \quad b = -3$

$c = \frac{(-3)^2}{4(5)}$

$c = \frac{9}{20}.$

$\therefore$ $\frac{9}{20}$ Must be added to $5x^2 - 3x$ to make it a perfect square expression.

6 (a) (i) $\frac{2y}{5} - \frac{y-2}{3} = \frac{y}{5}$

Multiplying by 15 each term we have

$6y - 5(y-2) = 3y$ ..... (00½)

$6y - 5y + 10 = 3y$

$y + 10 = 3y$ ..... (00½)

$2y = 10$

$y = 5$ ..... (01)

(ii) $-4x^2 + 8x - 3 = 0$

$a = -4 \quad b = 8 \quad c = -3.$

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ } ..... (01)

$= \frac{-8 \pm \sqrt{64 - 4(-4)(-3)}}{2(-4)}$

$= \frac{-8 \pm \sqrt{64 - 48}}{-8}$ ..... (01)

$= \frac{-8 \pm \sqrt{16}}{-8}$

$= \frac{-8 \pm 4}{-8}$ ..... (01)

$x = 1/2$ or $x = 3/2$.

6 (b) (i) $\frac{2\sqrt{7}}{\sqrt{10}-\sqrt{7}} \times \frac{\sqrt{10}+\sqrt{7}}{\sqrt{10}+\sqrt{7}}$ ----- 01

$= \frac{2\sqrt{70}+14}{10-7}$ ----- 00½

$= \frac{2\sqrt{70}+14}{3}$ ----- 00½

(ii) $5^x + 5^{x+1}$

$= 5^x + 5^x \cdot 5$ ---- 00½

$= 5^x(1+5)$

$= 5^x(6)$ ---- 01

$\therefore \frac{5^x(6)}{6} = \frac{30}{6}$ ----- 01

$5^x = 5^1$ ----- 01

$x = 1.$

7. (a) (i) $5670000 = 5.6 \times 10^6$ ----- (01)

(ii) $0.00024 = 2.4 \times 10^{-4}$ ---- (01)

b) (i) $2\log 5 + \log 36 - \log 9$

$= \log 25 + \log 36 - \log 9$ ------ (01)

$= \log\left(\frac{25 \times 36}{9}\right)$ } ------ (01)

$= \log 100$

$= \underline{\underline{2}}$ ------ (01)

(ii) $\log_3(\sqrt[3]{60}) = \frac{\log \sqrt[3]{60}}{\log 3}$ ------ (01)

$= \frac{\frac{1}{3}\log(10 \times 3 \times 2)}{\log 3.}$ --- (01)

$= \frac{\frac{1}{3}[\log 10 + \log 3 + \log 2)}{\log 3.}$ --- (01)

$= \frac{\frac{1}{3}(1 + 0.4771 + 0.310)}{0.4771}$ --- (01)

$= 1.24$ correct to 3 significant figures. --- (01)

8 (a) $\mathcal{U} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$

$A = \{2, 3, 5, 7, 11\}$ ---------- 01½

$B = \{1, 3, 5, 7, 9, 11\}$ ---------- 01½

(i) $A \cap B = \{3, 5, 7, 11\}$ ---------- 01

(ii) $A' \cap B'$

$A' = \{1, 4, 6, 8, 10\}$ ---------- 01

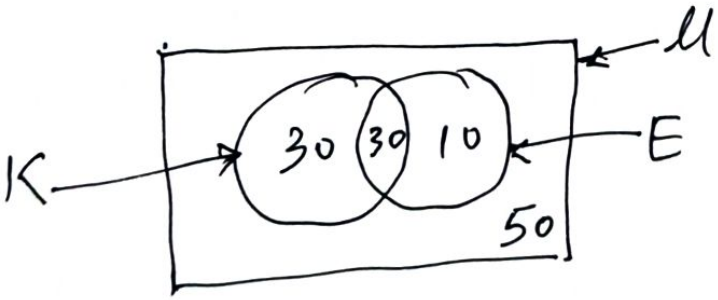
$B' = \{2, 4, 6, 8, 10\}$ ---------- 01

$A' \cap B' = \{4, 6, 8, 10\}$. ---------- 01

(b) let $\mathcal{U} = \{$All students in the school$\}$

$E = \{$All students studying English$\}$ ---- 01

$K = \{$All students studying Kiswa.$\}$



---------- 02

(i) English only = 10 students ---------- 01

(ii) Neither Eng nor Kisw = 50 Students. ---- 01

9 (a) from line (i)

$$-1 = \frac{y-3}{x+1} \quad \text{---} \quad 01$$

$$y - 3 = -x - 1$$

$$y = -x + 2 \quad \text{---} \quad 01$$

$x = 0, y = 2$; $y = 0, x = 2$

Points (0, 2) and (2, 0)

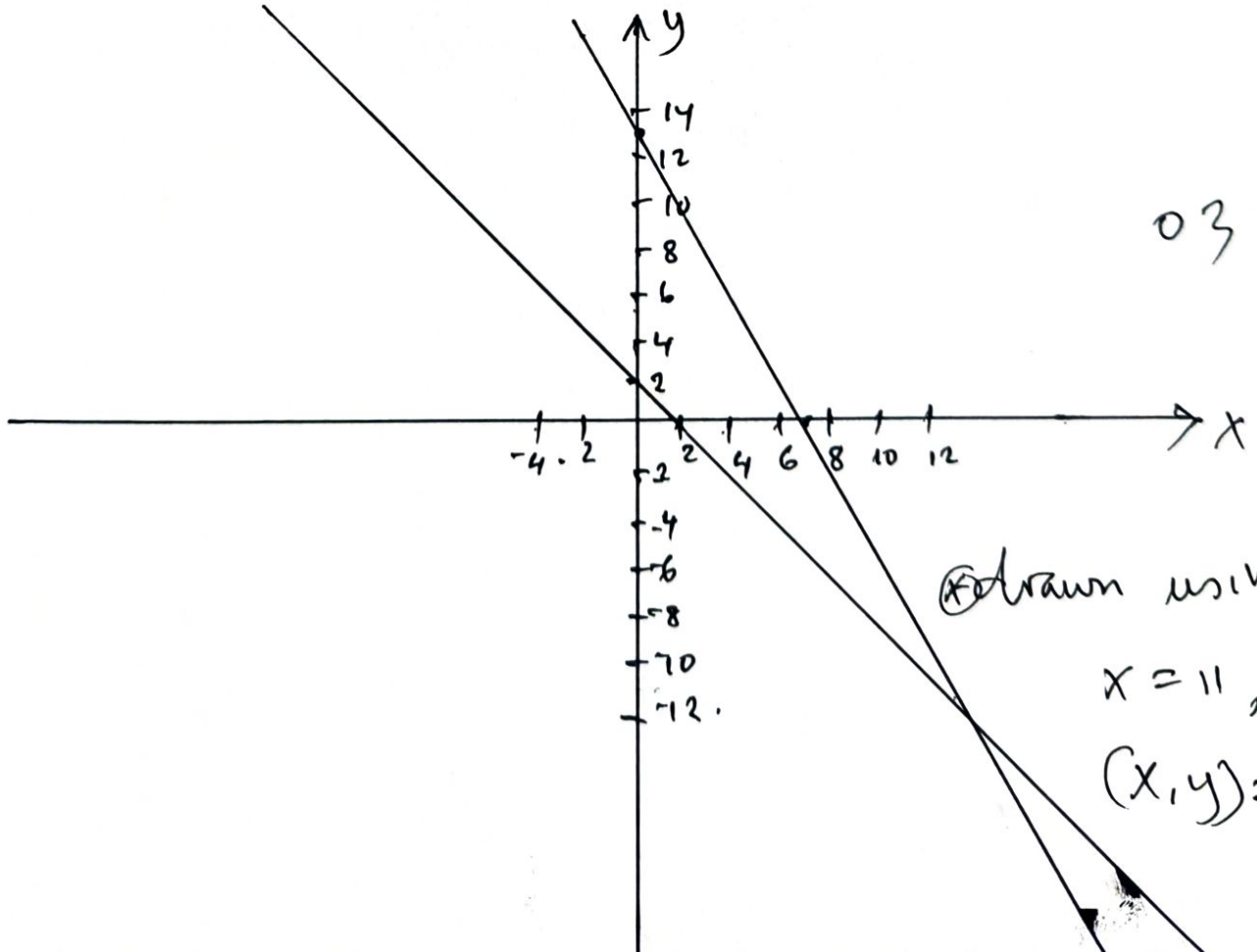
line (ii) $-2 = \frac{y-3}{x-5}$ --- 01

$$y - 3 = -2x + 10$$

$$y = -2x + 13 \quad \text{---} \quad 01$$

$x = 0, y = 13$, $y = 0, x = \frac{13}{2}$

Points (0, 13) and $(\frac{13}{2}, 0)$



03

Drawn using grap

$x = 11, y = -9$

$(x, y) = (11, -9)$

9 (b) (i) points from the graph

(0, 2) and (2, 0) 01/2

slope = $\frac{-2}{2}$

= -1. 01

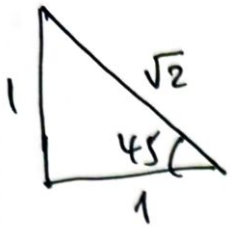
Eqn. $-1 = \frac{y-2}{x-0}$ 01/2

$y - 2 = -x + 0.$

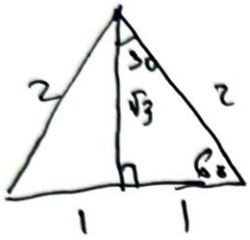
$y - 2 = -x$

$x + y - 2 = 0$. as required : . . . 01

10 (a) Using Special angles.



$\cos 45 = \frac{1}{\sqrt{2}}$ ——— 01



$\cos 30 = \frac{\sqrt{3}}{2}$

$\sin 30 = \frac{1}{2}$

$\sin 60 = \frac{\sqrt{3}}{2}$ ———— 01

$$\frac{\cos^2 45 - \sin^2 30}{\cos 30 + \sin 60} = \frac{\left(\frac{1}{\sqrt{2}}\right)^2 - \left(\frac{1}{2}\right)^2}{\frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2}}$$ ——— 01

$$= \frac{\frac{1}{2} - \frac{1}{4}}{\sqrt{3}}$$

$$= \frac{1}{4} \cdot \frac{1}{\sqrt{3}}$$

$$= \frac{1}{4\sqrt{3}}$$

$$= \frac{\sqrt{3}}{12}$$ ——— 01

10 a(iv) from $\triangle ABC$

$$\overline{AB}^2 = \overline{BC}^2 + AC^2$$

$$17^2 = \overline{BC}^2 + 8^2$$

$$17^2 - 8^2 = \overline{BC}^2$$

$$BC = \sqrt{17^2 - 8^2}$$

$$= 15\text{cm.} \quad \text{---} \quad 00\tfrac{1}{2}$$

$$\overline{DC} = \overline{BC} - \overline{BD}$$

$$= 15 - 9$$

$$= 6\text{cm.} \quad \text{---} \quad 00\tfrac{1}{2}$$

$$\cos X = \frac{\overline{DC}}{\overline{AC}}$$

$$= \frac{15}{17} \quad \text{---} \quad 00\tfrac{1}{2}$$

From $\triangle ADC$

$$\overline{AD}^2 = \overline{DC}^2 + \overline{AC}^2$$

$$AD = \sqrt{6^2 + 8^2}$$

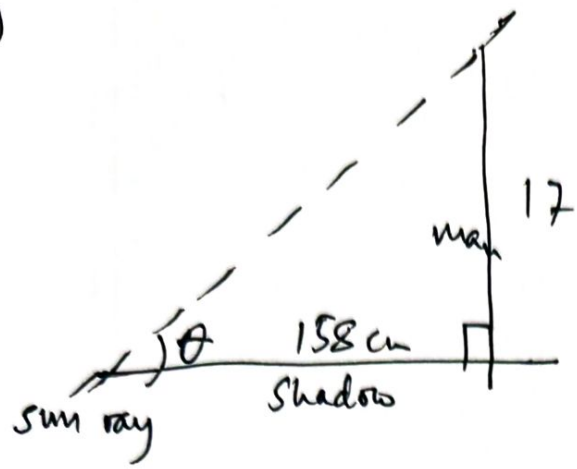
$$= 10\text{cm.}$$

$$\sin y = 6/10 = \frac{3}{5} \quad \text{---} \quad 00\tfrac{1}{2}$$

$$\cos X + \sin y = \frac{15}{17} + \frac{3}{5}$$

$$= \frac{126}{85} \quad \text{---} \quad 01$$

10. (b)



01

$$\tan\theta = \frac{172}{158}$$

$$= 1.089$$

01

$$\theta = \tan^{-1}(1.089)$$

$$= 47^\circ 26'$$

01

∴ Angle of elevation of the sun is $47^\circ 26'$